Department of Electronics & Telecommunication Engineering
Academic Year (2025-2026)

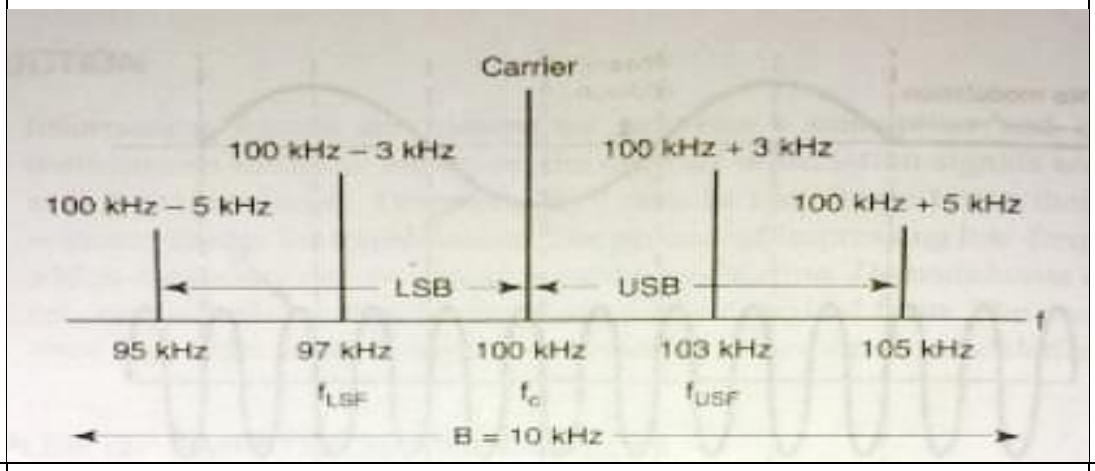
Subject: Principles of Communication Engineering **SY – V**

Module I ASSIGNMENT : Analog Transmission and Reception.

DATE OF ISSUE: 24/02/2026

CO MAPPING TO THIS ASSIGNMENT:	**CO1:** Identify Define AM and FM techniques, analyze them in time and frequency domains, and explain their generation and detection methods. Compare power requirements, bandwidth, and hardware complexity.

Q. No	Statement	Marks
1.	An audio frequency signal 20 sin 2π(500t) is used to amplitude modulate the carrier of 50 sin 2π(10^3 t) calculate. i) Modulation index, ii) Side band frequencies, iii) Amplitude of each side band iv) Band width v) Total power delivered to load of 600Ω	**5M**
2.	Draw and explain block diagram of super heterodyne receiver	**5M**
3.	One input to AM modulator is 500 KHZ carrier with an amplitude 20Vp. The second input is a 10KHZ signal with amplitude to cause a change in output wave of 7.5Vp. Determine (a) Upper and lower side frequencies (b) Modulation coefficient and percentage modulation (c) peak amplitude of the modulated carrier and upper and lower side frequency voltages. (d) maximum and minimum amplitude of the envelope. [Ans: 510 KHz, 490 KHz, 0.375, 37.5%3.75 V, 27.5 V, 12.5 V]	**5M**
4.	An AM wave is expressed as; 10(1+ 0.5cos 10^5 t + 0.2 cos 4000t) cos 10^7 t. List the different frequency components, [Ans: 15923.56 Hz, 636.95 Hz, 15.92 MHz]	**5M**
5.	For an AM modulator carrier frequency is 100 KHz and maximum modulating signal frequency is 5 KHz. Determine (a) frequency limits for the upper and lower side bands. (b) Band width (c) Upper and lower side frequencies produced when the modulating signal is a single frequency 3Khz tone. (d) draw the output frequency spectrum.	**5M**

	[Ans : 95 KHz to 100 KHz, 100 KHz to 105 Khz, 10KHz, 103 KHZ, 97 Khz]  	
6.	Explain AM transmitter with block diagram.	**5M**
7.	Explain Superhetrodyne receiver with block diagram	**5M**
8.	Explain Armstrong method for FM generation with block diagram.	**5M**
9.	Classify FM generation methods & with suitable diagram explain FM generation by Armstrong's Indirect method.	**5M**
10.	A frequency modulated signal is given by $x_c(t) = 10cos\{[2 \times 10^8 t] + sin[2\pi \times 200t]\}$, Determine: i) The carrier frequency. ii) Peak frequency deviation iii) The modulation Index	**5M**
11.	A frequency modulated signal is represented by voltage equation e_{FM}=10 sin[6×10_8t+5sin1250 t] calculate i) Carrier frequency f_c ii) Modulating frequency (FM) iii) Max_m deviation δ iv) What power will this FM wave dissipates in 20Ω resistor	**5M**
12.	FM wave is represented by following eqn V = 20 sin [108t + 4 sin 1200t] calculate, i) Carrier frequency ii) Modulating frequency iii) Modulation index & maximum deviation iv) Power dissipated by FM wave in 8Ω resistor.	**5M**
13.	A message signal m(t)=cos 2000πt+2cos 4000πt modulates the carrier c(t)=100 cos2πfct where fc=1MHz to produce the DSB signal. (a)	**5M**

	Determine the expression for USB signal (b)Determine and sketch the spectrum of USB signal.	
14.	An FM signal is given by $s(t) = 10\cos[2\pi(10^8)t + 5\sin(2\pi \cdot 10^3 t)]$. 1. Find the frequency deviation Δf. 2. Find the modulation index β. 3. Estimate the bandwidth using **Carson's Rule**.	**5M**
15.	A 25MHZ carrier is modulated by a 400Hz audio sine wave. If the carrier voltage is 4V and the maximum deviation is 10KHZ. Write down the voltage equation of the FM wave	**5M**
16.	A frequency modulated voltage is given by the equation $e = 12\cos(6 \times 10^8\ t + 5\sin 1250t)$ Find (1) Carrier frequency (2) Signal Frequency (3) Modulation Index (4) Maximum Frequency deviation (5) Power dissipated by the FM wave in 10 Ohm resistor	**6M**
17.	For an FM modulator where the modulating index, m_f =1, a modulating signal $e_m(t) = E_m \sin(2\pi\ 1000t)$, and an unmodulated carrier $e_c(t) = 10 \sin(2\pi\ 5 \times 10^5 t)$, determine: (a) the number of sets of significant side frequencies (b) the bandwidth of the FM signal (c) draw the amplitude spectrum (d) From Bessel Table, m_f = 1 yields a carrier component and 3 sets of significant side frequencies (n = 3).	**6M**
18.	Explain FM demodulation using PLL	**6M**
19.	For an AM DSBFC wave with a peak unmodulated carrier voltage Vc = 10 Vp, frequency of 100kHz, a load resistor of RL = 10 W, frequency of modulating signal of 10kHz and m = 1, determine the following i) Powers of the carrier and the upper and lower sidebands. ii) Total power of the modulated wave. iii) Bandwidth of the transmitted wave. iv) Draw the power and frequency spectrum.	**5M**
20.	The antenna current of an AM transmitter is 8A when only the carrier is sent but it increases to 8.93A when the carrier is sinusoidally modulated. Find the % age modulation.	**5M**